

GrounUp Enterprise Specification

PHASE 12

Enterprise Fleet, Equipment, Asset, Maintenance & Telematics Management

Version 1.0.0

Status: Review-ready

Proposed Lock Tag: GES-P12-v1.0.0

AI-Native | Multi-Tenant | Configurable | Mobile/Offline | API-First

Document Control

Control	Value
Specification ID	GES-P12-001
Phase	12
Version	1.0.0
Status	Review-ready
Proposed lock tag	GES-P12-v1.0.0
Owner	GrounUp Enterprise Architecture
Primary domain	Fleet, equipment, maintenance, telematics, fuel and asset lifecycle
Dependencies	Phases 01-11

Executive Summary

Phase 12 establishes the enterprise system of record and intelligence layer for owned, leased, rented, and customer-managed equipment, vehicles, tools, attachments, components, and supporting infrastructure. It unifies lifecycle, maintenance, inspection, telematics, fuel, dispatch, utilization, compliance, costing, warranty, and replacement decisions into a continuously updated Fleet Digital Twin.

The specification is designed for heavy civil, excavation, demolition, roadwork, utilities, vertical construction, landscaping, specialty trades, and mixed fleets. All capabilities are configurable rather than hard-coded and support tenant-specific terminology, workflows, thresholds, approvals, rate structures, asset classes, integrations, branding, and industry packs.

Governed Scope and Counts

Artifact	Count
Controlled requirements	280
Canonical entities	125
Data dictionary fields	180
Business rules	50
Validation rules	50
Workflows	26
Engines	32
API resources	40
UI surfaces	22
Reports	20
KPIs	36
Mapped tests	280

Architecture Principles

- Single governed fleet and asset identity across projects, finance, procurement, workforce, safety, and operations.
- Event-driven updates with effective dating, immutable audit history, optimistic concurrency, and tenant isolation.
- Configuration-first lifecycle states, maintenance policies, approval chains, inspection templates, rates, and notifications.
- Offline-capable field execution with deterministic synchronization and visible conflict resolution.
- AI recommendations are explainable, versioned, confidence-scored, evidence-linked, and subject to human approval for consequential decisions.
- Financial and project charge records reconcile with Phase 10 accounting and Phase 07 project operations.

Functional Domains

Fleet Registry

Fleet and equipment master records, ownership, classification, specifications, and lifecycle status. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Asset Lifecycle

Acquisition, commissioning, transfer, capitalization, utilization, retirement, disposal, and replacement. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Preventive Maintenance

Meter-, calendar-, condition-, and risk-based preventive maintenance programs. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Corrective Maintenance

Defect intake, diagnostics, work orders, labor, parts, repair execution, testing, and return to service. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Inspections

Pre-use, post-use, annual, regulatory, safety, and quality inspections with mobile evidence. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Telematics & IoT

Oem and third-party telematics ingestion, normalization, health events, location, meter, and utilization data. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Fuel & Energy

Fuel cards, bulk tanks, mobile fueling, electric charging, idle control, consumption, and reconciliation. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Parts & Components

Parts catalogs, serialized components, warranties, cores, inventory integration, and component history. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Work Orders

Planning, approval, scheduling, assignment, execution, cost capture, closeout, and audit. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Shop Operations

Shop bays, technician capacity, backlog, dispatch, tooling, calibration, and productivity. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Field Service

Service trucks, mobile technicians, emergency response, remote diagnostics, and site repair. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Rental & Leased Assets

Rental intake, rate terms, utilization, off-rent, damage, extensions, and invoice validation. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Compliance

Registrations, permits, dot records, emissions, inspections, certifications, and out-of-service controls. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Safety & Risk

Operator authorization, defect severity, lockout/tagout, incident linkage, and critical control verification. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Utilization & Dispatch

Availability, reservation, assignment, dispatch, location, shift use, and underutilization detection. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Equipment Costing

Ownership cost, operating cost, internal rates, fuel, maintenance, depreciation, and project chargeback. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Warranty & Claims

Coverage rules, claim preparation, evidence, recovery, denial tracking, and supplier reimbursement. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Tires & Undercarriage

Position tracking, measurements, rotations, replacements, wear forecasts, and cost-per-hour analysis. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Attachments & Tools

Attachment compatibility, custody, inspections, calibration, utilization, and loss prevention. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Documents & Media

Manuals, photos, invoices, inspection forms, service records, and immutable evidence. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Mobile & Offline

Offline inspections, work orders, parts usage, signatures, photos, sync, and conflict resolution. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Fleet Intelligence

Health scoring, failure prediction, replacement planning, cost anomaly detection, and recommendations. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Fleet Digital Twin

Live asset state combining physical, operational, maintenance, financial, location, and risk data. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Administration & Integration

Tenant configuration, white-label settings, rbac, apis, events, data migration, and audit. The domain supports configurable statuses, permissions, validations, notifications, audit evidence, APIs, reporting, mobile behavior, and cross-phase integration.

Canonical Workflows

P12-WF-01 - Asset Acquisition to Commissioning

Domain: Fleet Registry. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-02 - Asset Transfer

Domain: Asset Lifecycle. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-03 - Daily Pre-Use Inspection

Domain: Preventive Maintenance. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-04 - Defect to Corrective Work Order

Domain: Corrective Maintenance. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-05 - Preventive Maintenance Planning

Domain: Inspections. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-06 - Work Order Planning and Approval

Domain: Telematics & IoT. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-07 - Shop Work Execution

Domain: Fuel & Energy. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-08 - Field Service Dispatch

Domain: Parts & Components. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-09 - Emergency Breakdown Response

Domain: Work Orders. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-10 - Parts Issue and Return

Domain: Shop Operations. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-11 - Warranty Claim Recovery

Domain: Field Service. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-12 - Fuel Transaction Reconciliation

Domain: Rental & Leased Assets. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-13 - Rental On-Rent to Off-Rent

Domain: Compliance. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-14 - Telematics Event Processing

Domain: Safety & Risk. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-15 - Compliance Renewal

Domain: Utilization & Dispatch. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-16 - Operator Authorization

Domain: Equipment Costing. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-17 - Asset Reservation and Dispatch

Domain: Warranty & Claims. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-18 - Project Equipment Chargeback

Domain: Tires & Undercarriage. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-19 - Tire Lifecycle

Domain: Attachments & Tools. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-20 - Undercarriage Monitoring

Domain: Documents & Media. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-21 - Attachment Custody

Domain: Mobile & Offline. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-22 - Asset Replacement Planning

Domain: Fleet Intelligence. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-23 - Asset Disposal

Domain: Fleet Digital Twin. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-24 - Mobile Offline Sync

Domain: Administration & Integration. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-25 - Recall Campaign

Domain: Fleet Registry. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

P12-WF-26 - Digital Twin State Refresh

Domain: Asset Lifecycle. States: Initiated > Validated > Approved/Assigned > Executed > Verified > Closed. Controls: Configurable approval, notifications, SLA clocks, exception queues, and audit trail. Channels: Mobile/API/Web/Event.

Business and Validation Controls

Business controls prevent unauthorized, unsafe, financially invalid, or noncompliant actions. Validations are shared across UI, APIs, imports, integrations, and offline synchronization to prevent channel-specific exceptions.

ID	Domain	Control	Enforcement
P12-BR-001	Fleet Registry	Apply configurable tenant policy to fleet registry transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-002	Asset Lifecycle	Apply configurable tenant policy to asset lifecycle transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-003	Preventive Maintenance	Apply configurable tenant policy to preventive maintenance transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-004	Corrective Maintenance	Apply configurable tenant policy to corrective maintenance transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-005	Inspections	Apply configurable tenant policy to inspections transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-006	Telematics & IoT	Apply configurable tenant policy to telematics & iot transactions; block unsafe,	Blocking

		unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	
P12-BR-007	Fuel & Energy	Apply configurable tenant policy to fuel & energy transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-008	Parts & Components	Apply configurable tenant policy to parts & components transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-009	Work Orders	Apply configurable tenant policy to work orders transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-010	Shop Operations	Apply configurable tenant policy to shop operations transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-011	Field Service	Apply configurable tenant policy to field service transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-012	Rental & Leased Assets	Apply configurable tenant policy to rental & leased assets transactions; block unsafe, unauthorized,	Blocking

		duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	
P12-BR-013	Compliance	Apply configurable tenant policy to compliance transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-014	Safety & Risk	Apply configurable tenant policy to safety & risk transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-015	Utilization & Dispatch	Apply configurable tenant policy to utilization & dispatch transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-016	Equipment Costing	Apply configurable tenant policy to equipment costing transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-017	Warranty & Claims	Apply configurable tenant policy to warranty & claims transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking
P12-BR-018	Tires & Undercarriage	Apply configurable tenant policy to tires & undercarriage transactions; block unsafe, unauthorized, duplicate,	Blocking

		financially invalid, or noncompliant actions and require documented override authority for exceptions.	
P12-BR-019	Attachments & Tools	Apply configurable tenant policy to attachments & tools transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Advisory
P12-BR-020	Documents & Media	Apply configurable tenant policy to documents & media transactions; block unsafe, unauthorized, duplicate, financially invalid, or noncompliant actions and require documented override authority for exceptions.	Blocking

Engines and AI Governance

P12-ENG-01 - Asset Registry Engine

Deterministic and/or governed AI service for asset registry engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-02 - Lifecycle State Engine

Deterministic and/or governed AI service for lifecycle state engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-03 - Maintenance Due Engine

Deterministic and/or governed AI service for maintenance due engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-04 - Work Order Orchestration Engine

Deterministic and/or governed AI service for work order orchestration engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-05 - Inspection Rules Engine

Deterministic and/or governed AI service for inspection rules engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-06 - Defect Severity Engine

Deterministic and/or governed AI service for defect severity engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-07 - Telematics Normalization Engine

Deterministic and/or governed AI service for telematics normalization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-08 - Meter Integrity Engine

Deterministic and/or governed AI service for meter integrity engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-09 - Fuel Reconciliation Engine

Deterministic and/or governed AI service for fuel reconciliation engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-10 - Idle Analytics Engine

Deterministic and/or governed AI service for idle analytics engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-11 - Utilization Engine

Deterministic and/or governed AI service for utilization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-12 - Availability Engine

Deterministic and/or governed AI service for availability engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-13 - Dispatch Optimization Engine

Deterministic and/or governed AI service for dispatch optimization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-14 - Parts Demand Engine

Deterministic and/or governed AI service for parts demand engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-15 - Warranty Eligibility Engine

Deterministic and/or governed AI service for warranty eligibility engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-16 - Warranty Recovery Engine

Deterministic and/or governed AI service for warranty recovery engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-17 - Compliance Calendar Engine

Deterministic and/or governed AI service for compliance calendar engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-18 - Operator Authorization Engine

Deterministic and/or governed AI service for operator authorization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-19 - Equipment Cost Engine

Deterministic and/or governed AI service for equipment cost engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-20 - Internal Rate Engine

Deterministic and/or governed AI service for internal rate engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-21 - Project Chargeback Engine

Deterministic and/or governed AI service for project chargeback engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-22 - Replacement Planning Engine

Deterministic and/or governed AI service for replacement planning engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-23 - Failure Prediction Engine

Deterministic and/or governed AI service for failure prediction engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-24 - Asset Health Score Engine

Deterministic and/or governed AI service for asset health score engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-25 - Tire Wear Engine

Deterministic and/or governed AI service for tire wear engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-26 - Undercarriage Wear Engine

Deterministic and/or governed AI service for undercarriage wear engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-27 - Rental Optimization Engine

Deterministic and/or governed AI service for rental optimization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-28 - Geofence Event Engine

Deterministic and/or governed AI service for geofence event engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-29 - Alert Prioritization Engine

Deterministic and/or governed AI service for alert prioritization engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-30 - Mobile Sync Engine

Deterministic and/or governed AI service for mobile sync engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-31 - Fleet Digital Twin Engine

Deterministic and/or governed AI service for fleet digital twin engine. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

P12-ENG-32 - Fleet AI Copilot

Deterministic and/or governed AI service for fleet ai copilot. Human approval required for financial, safety, compliance, disposal, or vendor-impacting decisions. Inputs, outputs, model/rule version, confidence, overrides, and audit evidence retained.

Fleet Digital Twin

The Fleet Digital Twin is the governed, time-aware representation of each asset and the fleet as a whole. It combines identity and specifications, ownership and contracts, location, meters, telematics, diagnostics, inspections, defects, maintenance, labor, parts, fuel and energy, compliance, utilization, downtime, project assignments, cost, warranty, risk, health, replacement, documents, and decisions.

Twin state updates must be idempotent, source-attributed, time-stamped, versioned, and queryable as current state or historical snapshot. Conflicting observations are retained and resolved through source precedence, confidence, recency, and authorized human review.

API and Integration Standard

- OAuth2/OIDC, tenant context, least-privilege RBAC and service scopes.
- Idempotency keys for write operations and ETags for concurrency.
- Correlation IDs connecting source event, workflow, financial transaction, notification, and audit record.
- Webhooks and event streams for state changes, alerts, work orders, telematics, meter, compliance, and digital twin updates.
- Supported connector patterns for OEM telematics, fuel cards, GPS, ERP/accounting, procurement, payroll, project systems, identity, and document storage.

UI, Mobile and Offline Standard

Responsive web is the primary enterprise experience. Native/offline mobile supports pre-use inspections, defect reporting, work order execution, technician labor, parts and fluids, meter readings, photos, signatures, location capture, service truck stock, and return-to-service verification. Offline submissions remain encrypted, identity-bound, time-stamped, and conflict-aware.

Reporting and KPI Governance

KPI ID	KPI	Governed Definition
P12-KPI-01	Fleet Availability %	Governed calculation for fleet availability % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-02	Asset Utilization %	Governed calculation for asset utilization % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-03	Productive Hours %	Governed calculation for productive hours % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-04	Idle %	Governed calculation for idle % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-05	Planned Maintenance Compliance %	Governed calculation for planned maintenance compliance % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-06	PM On-Time Completion %	Governed calculation for pm on-time completion % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-07	Emergency Work %	Governed calculation for emergency work % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-08	Mean Time Between Failures	Governed calculation for mean time between failures using effective-dated fleet, work order, telematics, cost, project, and compliance data.

P12-KPI-09	Mean Time to Repair	Governed calculation for mean time to repair using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-10	Maintenance Cost per Hour	Governed calculation for maintenance cost per hour using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-11	Fuel Cost per Hour	Governed calculation for fuel cost per hour using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-12	Fuel Burn Variance	Governed calculation for fuel burn variance using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-13	Downtime Hours	Governed calculation for downtime hours using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-14	Work Order Backlog Hours	Governed calculation for work order backlog hours using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-15	Technician Utilization %	Governed calculation for technician utilization % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-16	First-Time Fix Rate	Governed calculation for first-time fix rate using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-17	Parts Fill Rate	Governed calculation for parts fill rate using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-18	Warranty Recovery \$	Governed calculation for warranty recovery \$ using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-19	Warranty Capture %	Governed calculation for warranty capture % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-20	Rental Utilization %	Governed calculation for rental utilization % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-21	Rental Leakage \$	Governed calculation for rental leakage \$ using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-22	Compliance On-Time %	Governed calculation for compliance on-time % using effective-dated fleet, work order, telematics, cost, project,

		and compliance data.
P12-KPI-23	Inspection Completion %	Governed calculation for inspection completion % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-24	Critical Defect Closure Time	Governed calculation for critical defect closure time using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-25	Operator Authorization Compliance %	Governed calculation for operator authorization compliance % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-26	Tire Cost per Hour	Governed calculation for tire cost per hour using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-27	Undercarriage Remaining Life	Governed calculation for undercarriage remaining life using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-28	Total Cost of Ownership	Governed calculation for total cost of ownership using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-29	Replacement Priority Index	Governed calculation for replacement priority index using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-30	Asset Health Score	Governed calculation for asset health score using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-31	Telematics Data Freshness %	Governed calculation for telematics data freshness % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-32	Meter Integrity %	Governed calculation for meter integrity % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-33	Project Charge Recovery %	Governed calculation for project charge recovery % using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-34	Fleet Carbon Intensity	Governed calculation for fleet carbon intensity using effective-dated fleet, work order, telematics, cost, project, and compliance data.
P12-KPI-35	AI Recommendation Acceptance %	Governed calculation for ai recommendation acceptance % using effective-dated fleet, work order, telematics, cost, project, and

		compliance data.
P12-KPI-36	Fleet Digital Twin Freshness	Governed calculation for fleet digital twin freshness using effective-dated fleet, work order, telematics, cost, project, and compliance data.

Testing and Traceability

Every controlled requirement maps to an entity, workflow, engine, API resource, and test case. Release gates require passing functional, calculation, authorization, tenant isolation, integration, offline sync, performance, security, audit, accessibility, AI governance, migration, and user acceptance tests.

Deployment and Cutover

1. Approve tenant configuration and asset taxonomy.
2. Cleanse and migrate asset, meter, maintenance, part, fuel, compliance, and cost histories.
3. Certify telematics, fuel, accounting, procurement, project, and identity integrations.
4. Validate roles, shops, technicians, mobile devices, offline policies, and inspection templates.
5. Pilot with a representative fleet and reconcile meters, work orders, fuel, cost, availability, and project charges.
6. Run parallel reporting and financial reconciliation before production lock.
7. Capture approvals, evidence, rollback plan, support ownership, and lock tag.

Approval and Lock Criteria

Phase 12 may be locked only after Product, Fleet Operations, Maintenance, Safety, Finance, Project Controls, Architecture, Security, Data Governance, Integration, Mobile, and UAT approvals are recorded; critical defects are closed; traceability is complete; and reconciliation evidence is accepted.

Appendix A - Requirement Register Sample

ID	Domain	Requirement	Test
P12-REQ-001	Fleet Registry	The platform shall create governed fleet registry records and decisions for fleet and equipment master records, ownership, classification, specifications, and lifecycle status, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-001
P12-REQ-002	Asset Lifecycle	The platform shall create governed asset lifecycle records and decisions for acquisition, commissioning, transfer, capitalization, utilization, retirement, disposal, and	P12-TC-002

		replacement, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	
P12-REQ-003	Preventive Maintenance	The platform shall create governed preventive maintenance records and decisions for meter-, calendar-, condition-, and risk-based preventive maintenance programs, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-003
P12-REQ-004	Corrective Maintenance	The platform shall create governed corrective maintenance records and decisions for defect intake, diagnostics, work orders, labor, parts, repair execution, testing, and return to service, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-004
P12-REQ-005	Inspections	The platform shall create governed inspections records and decisions for pre-use, post-use, annual, regulatory, safety, and quality inspections with mobile evidence, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-005
P12-REQ-006	Telematics & IoT	The platform shall create governed telematics & iot records and decisions for OEM and third-party telematics ingestion, normalization, health events, location, meter, and utilization data, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-006

P12-REQ-007	Fuel & Energy	The platform shall create governed fuel & energy records and decisions for fuel cards, bulk tanks, mobile fueling, electric charging, idle control, consumption, and reconciliation, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-007
P12-REQ-008	Parts & Components	The platform shall create governed parts & components records and decisions for parts catalogs, serialized components, warranties, cores, inventory integration, and component history, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-008
P12-REQ-009	Work Orders	The platform shall create governed work orders records and decisions for planning, approval, scheduling, assignment, execution, cost capture, closeout, and audit, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-009
P12-REQ-010	Shop Operations	The platform shall create governed shop operations records and decisions for shop bays, technician capacity, backlog, dispatch, tooling, calibration, and productivity, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-010
P12-REQ-011	Field Service	The platform shall create governed field service records and decisions for service trucks, mobile technicians, emergency	P12-TC-011

		response, remote diagnostics, and site repair, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	
P12-REQ-012	Rental & Leased Assets	The platform shall create governed rental & leased assets records and decisions for rental intake, rate terms, utilization, off-rent, damage, extensions, and invoice validation, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-012
P12-REQ-013	Compliance	The platform shall create governed compliance records and decisions for registrations, permits, DOT records, emissions, inspections, certifications, and out-of-service controls, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-013
P12-REQ-014	Safety & Risk	The platform shall create governed safety & risk records and decisions for operator authorization, defect severity, lockout/tagout, incident linkage, and critical control verification, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-014
P12-REQ-015	Utilization & Dispatch	The platform shall create governed utilization & dispatch records and decisions for availability, reservation, assignment, dispatch, location, shift use, and underutilization detection, with tenant isolation, role-based access, effective dating, audit history, configurable	P12-TC-015

		approvals, and API availability.	
P12-REQ-016	Equipment Costing	The platform shall create governed equipment costing records and decisions for ownership cost, operating cost, internal rates, fuel, maintenance, depreciation, and project chargeback, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-016
P12-REQ-017	Warranty & Claims	The platform shall create governed warranty & claims records and decisions for coverage rules, claim preparation, evidence, recovery, denial tracking, and supplier reimbursement, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-017
P12-REQ-018	Tires & Undercarriage	The platform shall create governed tires & undercarriage records and decisions for position tracking, measurements, rotations, replacements, wear forecasts, and cost-per-hour analysis, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-018
P12-REQ-019	Attachments & Tools	The platform shall create governed attachments & tools records and decisions for attachment compatibility, custody, inspections, calibration, utilization, and loss prevention, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-019
P12-REQ-020	Documents & Media	The platform shall create	P12-TC-020

		governed documents & media records and decisions for manuals, photos, invoices, inspection forms, service records, and immutable evidence, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	
P12-REQ-021	Mobile & Offline	The platform shall create governed mobile & offline records and decisions for offline inspections, work orders, parts usage, signatures, photos, sync, and conflict resolution, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-021
P12-REQ-022	Fleet Intelligence	The platform shall create governed fleet intelligence records and decisions for health scoring, failure prediction, replacement planning, cost anomaly detection, and recommendations, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-022
P12-REQ-023	Fleet Digital Twin	The platform shall create governed fleet digital twin records and decisions for live asset state combining physical, operational, maintenance, financial, location, and risk data, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-023
P12-REQ-024	Administration & Integration	The platform shall create governed administration & integration records and decisions for tenant configuration, white-label settings, RBAC, APIs, events, data migration,	P12-TC-024

		and audit, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	
P12-REQ-025	Fleet Registry	The platform shall update governed fleet registry records and decisions for fleet and equipment master records, ownership, classification, specifications, and lifecycle status, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-025
P12-REQ-026	Asset Lifecycle	The platform shall update governed asset lifecycle records and decisions for acquisition, commissioning, transfer, capitalization, utilization, retirement, disposal, and replacement, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-026
P12-REQ-027	Preventive Maintenance	The platform shall update governed preventive maintenance records and decisions for meter-, calendar-, condition-, and risk-based preventive maintenance programs, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-027
P12-REQ-028	Corrective Maintenance	The platform shall update governed corrective maintenance records and decisions for defect intake, diagnostics, work orders, labor, parts, repair execution, testing, and return to service, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API	P12-TC-028

		availability.	
P12-REQ-029	Inspections	The platform shall update governed inspections records and decisions for pre-use, post-use, annual, regulatory, safety, and quality inspections with mobile evidence, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-029
P12-REQ-030	Telematics & IoT	The platform shall update governed telematics & iot records and decisions for OEM and third-party telematics ingestion, normalization, health events, location, meter, and utilization data, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-030
P12-REQ-031	Fuel & Energy	The platform shall update governed fuel & energy records and decisions for fuel cards, bulk tanks, mobile fueling, electric charging, idle control, consumption, and reconciliation, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-031
P12-REQ-032	Parts & Components	The platform shall update governed parts & components records and decisions for parts catalogs, serialized components, warranties, cores, inventory integration, and component history, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-032
P12-REQ-033	Work Orders	The platform shall update governed work orders records and decisions for	P12-TC-033

		planning, approval, scheduling, assignment, execution, cost capture, closeout, and audit, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	
P12-REQ-034	Shop Operations	The platform shall update governed shop operations records and decisions for shop bays, technician capacity, backlog, dispatch, tooling, calibration, and productivity, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-034
P12-REQ-035	Field Service	The platform shall update governed field service records and decisions for service trucks, mobile technicians, emergency response, remote diagnostics, and site repair, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-035
P12-REQ-036	Rental & Leased Assets	The platform shall update governed rental & leased assets records and decisions for rental intake, rate terms, utilization, off-rent, damage, extensions, and invoice validation, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-036
P12-REQ-037	Compliance	The platform shall update governed compliance records and decisions for registrations, permits, DOT records, emissions, inspections, certifications, and out-of-service controls, with tenant isolation, role-based access, effective dating, audit history,	P12-TC-037

		configurable approvals, and API availability.	
P12-REQ-038	Safety & Risk	The platform shall update governed safety & risk records and decisions for operator authorization, defect severity, lockout/tagout, incident linkage, and critical control verification, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-038
P12-REQ-039	Utilization & Dispatch	The platform shall update governed utilization & dispatch records and decisions for availability, reservation, assignment, dispatch, location, shift use, and underutilization detection, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-039
P12-REQ-040	Equipment Costing	The platform shall update governed equipment costing records and decisions for ownership cost, operating cost, internal rates, fuel, maintenance, depreciation, and project chargeback, with tenant isolation, role-based access, effective dating, audit history, configurable approvals, and API availability.	P12-TC-040

Appendix B - Canonical Entity Register

ID	Entity	Definition
P12-ENT-001	Tenant	Canonical tenant record supporting governed fleet and asset operations.
P12-ENT-002	Legal Entity	Canonical legal entity record supporting governed fleet and asset operations.
P12-ENT-003	Business Unit	Canonical business unit record supporting governed fleet and asset operations.
P12-ENT-004	Fleet Asset	Canonical fleet asset record

		supporting governed fleet and asset operations.
P12-ENT-005	Asset Class	Canonical asset class record supporting governed fleet and asset operations.
P12-ENT-006	Asset Type	Canonical asset type record supporting governed fleet and asset operations.
P12-ENT-007	Asset Model	Canonical asset model record supporting governed fleet and asset operations.
P12-ENT-008	Manufacturer	Canonical manufacturer record supporting governed fleet and asset operations.
P12-ENT-009	Dealer	Canonical dealer record supporting governed fleet and asset operations.
P12-ENT-010	Ownership Record	Canonical ownership record record supporting governed fleet and asset operations.
P12-ENT-011	Lease Agreement	Canonical lease agreement record supporting governed fleet and asset operations.
P12-ENT-012	Rental Agreement	Canonical rental agreement record supporting governed fleet and asset operations.
P12-ENT-013	Asset Status History	Canonical asset status history record supporting governed fleet and asset operations.
P12-ENT-014	Asset Location	Canonical asset location record supporting governed fleet and asset operations.
P12-ENT-015	Geofence	Canonical geofence record supporting governed fleet and asset operations.
P12-ENT-016	Meter	Canonical meter record supporting governed fleet and asset operations.
P12-ENT-017	Meter Reading	Canonical meter reading record supporting governed fleet and asset operations.
P12-ENT-018	Telematics Device	Canonical telematics device record supporting governed fleet and asset operations.
P12-ENT-019	Telematics Provider	Canonical telematics provider record supporting governed fleet and asset operations.
P12-ENT-020	Telematics Event	Canonical telematics event record supporting governed fleet and asset operations.
P12-ENT-021	Diagnostic Trouble Code	Canonical diagnostic trouble code record supporting governed fleet and asset operations.
P12-ENT-022	Engine Hour Reading	Canonical engine hour reading record supporting governed fleet and asset operations.
P12-ENT-023	Odometer Reading	Canonical odometer reading record supporting governed fleet and asset

		operations.
P12-ENT-024	Fuel Transaction	Canonical fuel transaction record supporting governed fleet and asset operations.
P12-ENT-025	Fuel Card	Canonical fuel card record supporting governed fleet and asset operations.
P12-ENT-026	Fuel Tank	Canonical fuel tank record supporting governed fleet and asset operations.
P12-ENT-027	Fuel Delivery	Canonical fuel delivery record supporting governed fleet and asset operations.
P12-ENT-028	Charging Session	Canonical charging session record supporting governed fleet and asset operations.
P12-ENT-029	Energy Tariff	Canonical energy tariff record supporting governed fleet and asset operations.
P12-ENT-030	Work Order	Canonical work order record supporting governed fleet and asset operations.
P12-ENT-031	Work Order Task	Canonical work order task record supporting governed fleet and asset operations.
P12-ENT-032	Maintenance Program	Canonical maintenance program record supporting governed fleet and asset operations.
P12-ENT-033	Maintenance Trigger	Canonical maintenance trigger record supporting governed fleet and asset operations.
P12-ENT-034	Maintenance Schedule	Canonical maintenance schedule record supporting governed fleet and asset operations.
P12-ENT-035	Service Interval	Canonical service interval record supporting governed fleet and asset operations.
P12-ENT-036	Defect	Canonical defect record supporting governed fleet and asset operations.
P12-ENT-037	Failure	Canonical failure record supporting governed fleet and asset operations.
P12-ENT-038	Diagnostic Record	Canonical diagnostic record record supporting governed fleet and asset operations.
P12-ENT-039	Repair Action	Canonical repair action record supporting governed fleet and asset operations.
P12-ENT-040	Technician	Canonical technician record supporting governed fleet and asset operations.
P12-ENT-041	Technician Skill	Canonical technician skill record supporting governed fleet and asset operations.
P12-ENT-042	Shop	Canonical shop record supporting governed fleet and asset operations.

P12-ENT-043	Shop Bay	Canonical shop bay record supporting governed fleet and asset operations.
P12-ENT-044	Service Truck	Canonical service truck record supporting governed fleet and asset operations.
P12-ENT-045	Labor Entry	Canonical labor entry record supporting governed fleet and asset operations.
P12-ENT-046	Part	Canonical part record supporting governed fleet and asset operations.
P12-ENT-047	Part Inventory Link	Canonical part inventory link record supporting governed fleet and asset operations.
P12-ENT-048	Part Issue	Canonical part issue record supporting governed fleet and asset operations.
P12-ENT-049	Core Return	Canonical core return record supporting governed fleet and asset operations.
P12-ENT-050	Component	Canonical component record supporting governed fleet and asset operations.
P12-ENT-051	Component Position	Canonical component position record supporting governed fleet and asset operations.
P12-ENT-052	Component Installation	Canonical component installation record supporting governed fleet and asset operations.
P12-ENT-053	Warranty	Canonical warranty record supporting governed fleet and asset operations.
P12-ENT-054	Warranty Claim	Canonical warranty claim record supporting governed fleet and asset operations.
P12-ENT-055	Inspection Template	Canonical inspection template record supporting governed fleet and asset operations.
P12-ENT-056	Inspection	Canonical inspection record supporting governed fleet and asset operations.
P12-ENT-057	Inspection Item	Canonical inspection item record supporting governed fleet and asset operations.
P12-ENT-058	Inspection Finding	Canonical inspection finding record supporting governed fleet and asset operations.
P12-ENT-059	Corrective Action	Canonical corrective action record supporting governed fleet and asset operations.
P12-ENT-060	Lockout Record	Canonical lockout record record supporting governed fleet and asset operations.
P12-ENT-061	Operator Authorization	Canonical operator authorization record supporting governed fleet and asset operations.

P12-ENT-062	Asset Assignment	Canonical asset assignment record supporting governed fleet and asset operations.
P12-ENT-063	Reservation	Canonical reservation record supporting governed fleet and asset operations.
P12-ENT-064	Dispatch	Canonical dispatch record supporting governed fleet and asset operations.
P12-ENT-065	Utilization Record	Canonical utilization record record supporting governed fleet and asset operations.
P12-ENT-066	Idle Event	Canonical idle event record supporting governed fleet and asset operations.
P12-ENT-067	Downtime Event	Canonical downtime event record supporting governed fleet and asset operations.
P12-ENT-068	Availability Calendar	Canonical availability calendar record supporting governed fleet and asset operations.
P12-ENT-069	Rate Schedule	Canonical rate schedule record supporting governed fleet and asset operations.
P12-ENT-070	Internal Rental Rate	Canonical internal rental rate record supporting governed fleet and asset operations.
P12-ENT-071	Cost Ledger	Canonical cost ledger record supporting governed fleet and asset operations.
P12-ENT-072	Depreciation Profile	Canonical depreciation profile record supporting governed fleet and asset operations.
P12-ENT-073	Replacement Plan	Canonical replacement plan record supporting governed fleet and asset operations.
P12-ENT-074	Capital Request	Canonical capital request record supporting governed fleet and asset operations.
P12-ENT-075	Disposal Record	Canonical disposal record record supporting governed fleet and asset operations.
P12-ENT-076	Auction Record	Canonical auction record record supporting governed fleet and asset operations.
P12-ENT-077	Registration	Canonical registration record supporting governed fleet and asset operations.
P12-ENT-078	Permit	Canonical permit record supporting governed fleet and asset operations.
P12-ENT-079	Compliance Requirement	Canonical compliance requirement record supporting governed fleet and asset operations.
P12-ENT-080	Compliance Evidence	Canonical compliance evidence record supporting governed fleet and asset operations.

P12-ENT-081	Emissions Record	Canonical emissions record record supporting governed fleet and asset operations.
P12-ENT-082	DOT Inspection	Canonical dot inspection record supporting governed fleet and asset operations.
P12-ENT-083	Tire	Canonical tire record supporting governed fleet and asset operations.
P12-ENT-084	Tire Position	Canonical tire position record supporting governed fleet and asset operations.
P12-ENT-085	Tire Measurement	Canonical tire measurement record supporting governed fleet and asset operations.
P12-ENT-086	Undercarriage Measurement	Canonical undercarriage measurement record supporting governed fleet and asset operations.
P12-ENT-087	Attachment	Canonical attachment record supporting governed fleet and asset operations.
P12-ENT-088	Tool	Canonical tool record supporting governed fleet and asset operations.
P12-ENT-089	Calibration Record	Canonical calibration record record supporting governed fleet and asset operations.
P12-ENT-090	Custody Record	Canonical custody record record supporting governed fleet and asset operations.
P12-ENT-091	Accident Link	Canonical accident link record supporting governed fleet and asset operations.
P12-ENT-092	Safety Incident Link	Canonical safety incident link record supporting governed fleet and asset operations.
P12-ENT-093	Insurance Policy	Canonical insurance policy record supporting governed fleet and asset operations.
P12-ENT-094	Insurance Claim	Canonical insurance claim record supporting governed fleet and asset operations.
P12-ENT-095	Document	Canonical document record supporting governed fleet and asset operations.
P12-ENT-096	Photo	Canonical photo record supporting governed fleet and asset operations.
P12-ENT-097	Signature	Canonical signature record supporting governed fleet and asset operations.
P12-ENT-098	Approval	Canonical approval record supporting governed fleet and asset operations.
P12-ENT-099	Notification	Canonical notification record supporting governed fleet and asset operations.
P12-ENT-100	Alert	Canonical alert record supporting governed fleet and asset operations.

P12-ENT-101	Health Score	Canonical health score record supporting governed fleet and asset operations.
P12-ENT-102	Risk Score	Canonical risk score record supporting governed fleet and asset operations.
P12-ENT-103	Failure Prediction	Canonical failure prediction record supporting governed fleet and asset operations.
P12-ENT-104	Recommendation	Canonical recommendation record supporting governed fleet and asset operations.
P12-ENT-105	Digital Twin Snapshot	Canonical digital twin snapshot record supporting governed fleet and asset operations.
P12-ENT-106	Digital Twin Event	Canonical digital twin event record supporting governed fleet and asset operations.
P12-ENT-107	Integration Endpoint	Canonical integration endpoint record supporting governed fleet and asset operations.
P12-ENT-108	Integration Mapping	Canonical integration mapping record supporting governed fleet and asset operations.
P12-ENT-109	Import Batch	Canonical import batch record supporting governed fleet and asset operations.
P12-ENT-110	Export Batch	Canonical export batch record supporting governed fleet and asset operations.
P12-ENT-111	Audit Event	Canonical audit event record supporting governed fleet and asset operations.
P12-ENT-112	Role	Canonical role record supporting governed fleet and asset operations.
P12-ENT-113	Permission	Canonical permission record supporting governed fleet and asset operations.
P12-ENT-114	Configuration	Canonical configuration record supporting governed fleet and asset operations.
P12-ENT-115	Custom Field	Canonical custom field record supporting governed fleet and asset operations.
P12-ENT-116	Mobile Sync Batch	Canonical mobile sync batch record supporting governed fleet and asset operations.
P12-ENT-117	Conflict Record	Canonical conflict record record supporting governed fleet and asset operations.
P12-ENT-118	Vendor Link	Canonical vendor link record supporting governed fleet and asset operations.
P12-ENT-119	Purchase Order Link	Canonical purchase order link record supporting governed fleet and asset operations.

P12-ENT-120	Invoice Link	Canonical invoice link record supporting governed fleet and asset operations.
P12-ENT-121	Project Charge	Canonical project charge record supporting governed fleet and asset operations.
P12-ENT-122	Project Assignment	Canonical project assignment record supporting governed fleet and asset operations.
P12-ENT-123	Crew Assignment	Canonical crew assignment record supporting governed fleet and asset operations.
P12-ENT-124	Jobsite	Canonical jobsite record supporting governed fleet and asset operations.
P12-ENT-125	Weather Observation	Canonical weather observation record supporting governed fleet and asset operations.